

# Divergent Transpacific Corridors

## *Post-COVID U.S. Capacity Redistribution and the Commercial Case for Fukuoka Expansion*

Drew Dickson

June 2026

Data: BTS – T-100 International Segment (2019-2025), Form 41 Schedules P-12, P-12A, P-52, T-1, T-2. Q1 2026 Carrier Earnings releases

### **Abstract**

The transpacific is not simply one macro-level market. Between 2019 and 2025, five U.S.-Asia corridors diverged from one another sharply: China saw a structural collapse driven by bilateral frequency caps and regulation, recovering to just 29.1% of pre-pandemic passenger volumes. Japan rebounded to 92.5% of 2019 passengers, while operating above 2019 load factors – a corridor constrained by supply where available seats are filling faster than they did before COVID. Taiwan overshot this recovery, reaching 126% of 2019 passengers with seats at 142%, producing a load factor slide to 74%. South Korea normalized and has grown, reaching 111% of its 2019 passengers. Hong Kong recovered in the partial sense, with leisure demand restored but business traffic remaining fragmented. Overlaid on these corridor stories is a divergence in the performance of the three major U.S. carriers: United is the only U.S. legacy carrier growing Japan capacity, Delta pivoted to Korea through its Korean Air joint venture, and American has largely stagnated across the transpacific, without a clear strategic replacement or overhaul. This paper quantifies the dynamics using BTS T-100 International Segment data (2019-2025), and Form 41 cost schedules, then applies a calibrated route economics model, grounded by Q1 2026 carrier earnings, reported RASM and CASM values, and aircraft-level P-52 cost data, to evaluate new route candidates for U.S. nonstop service to Fukuoka, Japan's fourth-largest metro area and a growing business center.

## **Introduction**

Between 2020 and late 2022, U.S.-Asia transpacific service was largely suspended, reducing some of its corridors to freight contractors and diplomatic travel. When demand recovered in the years following, such recovery was anything but uniform. Japanese restrictions lifted later than Korean ones, Chinese negotiations heavily constrained supply independently of any demand shift. Taiwan accumulated capacity as carriers hedged against geopolitical risks. Hong Kong's evolution changed the nature of its travel base. Each independent corridor has moved on its own timeline, with differing competitive structures and carriers who had spent two years reworking their fleet assignments while also renegotiating joint venture terms.

The result, visible in BTS T-100 data through 2025, is a region of the transpacific network that appears similar to 2019 in aggregate terms but is structurally altered in many metrics that matter to network planners: load factor, carrier footprints, and gateway concentrations amongst others.

This paper has two main purposes: to document that reorganization of the network, and disaggregate the major corridors and describe, with data, what shifts occurred and why. Additionally, the paper aims to apply the financial framework that appears from that analysis to a specific network case study: is Fukuoka, Japan's fourth-largest metro and one of the few Japanese major gateways with no current nonstop U.S. service, commercially viable for new route entry?

The analysis draws on six years of T-100 data covering 2019-2025, Form 41 schedules P-12, P-12A, P-52, T-1, T-2, and Q1 2026 financial statements from United, Delta, and American. Where the data provides its findings, assumptions are stated accordingly so that sensitivity of said conclusions to those assumptions can be evaluated.

## **Data and Methodology**

*BTS T-100 International Segment (2019-2025)* – The International Segment database records scheduled service between U.S. and foreign points by carrier, O&D, and month. Fields used include total departures performed, available seats, and passenger data, allowing for the calculation of load factor rates. Five destination markets were observed: Japan, South Korea, Taiwan, Hong Kong, and China. All analysis is directional: U.S. origin to foreign destination.

*Form 41 Schedule P-52 (Aircraft Operating Expenses)* – P-52 records aircraft-level expense of operations by carrier, aircraft type, and year. Fields extracted include the total air operating expenses and direct maintenance costs. This schedule is the source for non-fuel CASM figures used in the route simulation; its granularity at the aircraft level allows for the isolation of widebody economics from the recorded system averages.

*Form 41 Schedule P-12A (Fuel)* – Quarterly fuel cost and consumption by carrier. Total cost and total gallons are recorded and used to derive fuel CASM against system ASMs recorded in Schedule T-2. As a validation metric, cost per gallon is computed.

*Form 41 Schedule P-12 (Carrier P&L)* – Annual carrier-level income statement data. Values are in thousands of dollars. Combined with T-2 capacity data, P-12 produces system-level yields in cents per revenue passenger mile and RASM in cents per available seat mile.

*Form 41 Schedule T-2 (System Capacity)* – Quarterly ASMs and RPMs by carrier. Used as the denominator for system-level CASM, yield, and RASM calculations. Aircraft type detail present in T-2 allows for fleet-level joining when combined with P-52. Note: T-2 contains three rows per carrier-quarter-aircraft combination under service class codes F, K, and Z with identical ASM and RPM values. Summing all rows inflates every figure by exactly 3x. All T-2 aggregations in this analysis supply the SERVICE\_CLASS = “F” filter before summarizing, returning the correct scheduled-service totals. This correction affects the reported non-fuel CASM figures relative to uncorrected findings.

*Form 41 Schedule T-1 (Regional Capacity)* – Carrier capacity which is broken down by region and service class. Pacific region ASMs, RPMs, and passenger counts are extracted for 2023-2025 to benchmark Pacific load factors against the overall system averages.

*Q1 2026 Earnings Releases* – United Airlines, Delta Air Lines, and American Airlines each publish statistical tables sharing their passenger amounts, load factor, yield, and PRASM. Aggregation levels vary: Delta reports solely their aggregated RASM while American reports it down to the region, allowing for a simple cross-reference against American’s reported metrics.

## **Corridor Recovery Framework**

All corridor statistics are indexed against 2019 as the pre-COVID baseline, where passenger recovery is defined as  $(\text{year } t \text{ passengers} / 2019 \text{ passengers}) * 100$ , with the same layout applied to seats. Load factor is calculated directly as passengers divided by seats for a given year.

There is an important distinction between passenger recovery and seat recovery. A corridor with 90% recovery with seats at 80% of 2019 is constrained on the supply-side; demand restoration has outpaced capacity restoration. Meanwhile, another corridor at 90% recovery with seats at 110% is demand-constrained; capacity growth has overshoot the demand recovery. These two situations give opposite conclusions about route viability, and the trajectory of the yields, yet they produce nearly identical passenger recovery amounts. Load factor – the rate of which seats are filled – is the metric that truly splits them apart.

## **Unit Economics Framework**

The simulation of routes requires three cost inputs: fuel CASM, non-fuel CASM, and an indirect cost multiplier.

Fuel CASM is sourced from Q1 2026 earnings releases: 3.91 cents/ASM for United, 3.96 cents for Delta (unadjusted for refinery benefit; 3.75 cents adjusted), and an estimated 3.91 cents for American based on published gallon costs. These figures reflect Q1 2026 fuel prices, which are

somewhat elevated relative to the 2025 full-year average of approximately 3.06–3.09 cents/ASM reflected in Form 41 annual data. Q1 2026 figures are used throughout the simulation for currency.

Non-fuel CASM is sourced from Form 41 P-52, at the aircraft level, with T-2 ASM denominators corrected via the SERVICE\_CLASS filter described above. P-52 covers direct aircraft operating costs: flying operations (crew), direct maintenance, and depreciation of flight equipment, and does not include station costs, passenger services, sales, or corporate overhead. These corrected figures produce: 7.11 cents per ASM for the United Boeing 787-9, 6.66 cents for the Delta Airbus A350-900, and 9.39 cents for the American Boeing 787-9. System CASM-ex figures reported in earnings releases (approximately 13–15 cents/ASM) are not appropriate for single-route transpacific costing because they blend narrow-body domestic operations with widebody international; the P-52 direct figures are the appropriate input.

The indirect operating cost (IOC) multiplier of 1.3x is applied to direct operating costs (fuel plus non-fuel P-52) to approximate total trip cost, covering the indirect costs excluded from P-52. Three fuel scenarios are presented: Base (Q1 2026 actual), Moderate (+20%), and Severe (+65%). The severe scenario acts as a proxy for a major geopolitical shock or supply disruption.

### **IOC Validation and SFO-KIX Context**

SFO-KIX: United's existing secondary Japan gateway, carrying 78,175 passengers on 352 departures in 2025 at an 80.5% load factor, provides structural context for Fukuoka. As a threshold observation: applying the Q1 2026 Pacific average yield of 16.77 cents/RPM, the model produces annual revenue of approximately \$70.1 million against annual costs of \$73.5 million, a margin of -4.8%. This does not necessarily indicate SFO-KIX is unprofitable; it indicates the route must earn above the Pacific average to be viable. A mature monopoly gateway with no U.S. competitor and strong ANA joint venture connecting traffic can plausibly earn that premium. The cost structure implies the premium is approximately 10-14% above the Pacific average, though the actual route-level yield of SFO-KIX is not observable in any public dataset.

This 10–14% premium range informs the plausibility framing in the yield sensitivity analysis that follows. The breakeven yield for Fukuoka, derived directly from the cost model, falls within that range for United and below it for Delta, making the corridor commercially reachable at maturity.

### **Revenue Assumptions, Secondary Market Yield**

Carrier Pacific yield from Q1 2026 earnings: 16.77 cents/RPM for United, 15.95 cents for Delta (indexed proxy, see Limitations), and 15.28 cents for American, represents the corridor average for each carrier across all Pacific routes operated. These are the baseline inputs to the simulation. No yield discount is applied below the Pacific average; the cost model shows that the Pacific average yield is itself insufficient to cover costs at 1.3x IOC.

The relevant question is therefore not whether Fukuoka earns a discount to the Pacific average, but whether it can eventually earn a premium above it. SFO-KIX's operating context implies a premium of approximately 10-14% is achievable for a mature U.S. secondary Japan monopoly gateway, though the route-level yield is not directly observed in any public dataset. Yield sensitivity is tested at three scenarios: Pacific average (0%), modest premium (+8%), and mature gateway premium (+14%). The breakeven yield for United's west-coast Fukuoka routes is approximately 18.6 cents/RPM, about +11% above United's Pacific average, within the range that the SFO-KIX cost analysis suggests is achievable at maturity.

## **Routing Adjustments**

On transpacific routes, aircraft will generally not fly the great circle route. ATC constraints and jet stream considerations produce track mile amounts that exceed the great circle stage length used to price tickets. A standard multiplier of 1.06x is applied to Pacific coast departures operating standard transpacific routing. East coast and other departure regions that must navigate around Russian airspace receive a 1.12x multiplier, consistent with the block time penalty U.S. carriers have reported to the DOT for routing that must avoid the airspace following geopolitical tensions.

Costs are calculated on track miles as fuel burn and crew time are a function of the actual distance flown. Revenue itself is calculated on great circle miles as passenger fares are priced on stage length. This creates a cost disadvantage for routes who are more severely impacted by circuitry, particularly ORD and New York departures to Asia.

## **Corridor Analysis**

Before examining each corridor in detail, it is necessary to view the aggregate picture. Five markets that entered 2019 at similar load factors – all within the 80-83% range – ended 2025 at load factors ranging from 74-84%, with passenger recovery ranging from 29-126%. The story resides in the variance amongst the corridors.

| <b>Corridor</b> | <b>2025 Pax</b> | <b>Pax Recovery</b> | <b>2025 LF</b> | <b>2019 LF</b> | <b>LF Change</b> | <b>Signal</b>       |
|-----------------|-----------------|---------------------|----------------|----------------|------------------|---------------------|
| China           | 1,236,200       | 29.1%               | 84.3%          | 81.4%          | +2.9 pp          | Structural collapse |
| Hong Kong       | 1,379,715       | 76.6%               | 83.5%          | 80.1%          | +3.4 pp          | Bifurcated recovery |
| Japan           | 4,958,188       | 92.5%               | 83.2%          | 81.4%          | +1.8 pp          | Supply-constrained  |

| Corridor    | 2025 Pax  | Pax Recovery | 2025 LF | 2019 LF | LF Change | Signal             |
|-------------|-----------|--------------|---------|---------|-----------|--------------------|
| South Korea | 3,319,914 | 111.0%       | 80.1%   | 83.3%   | -3.2 pp   | Full normalization |
| Taiwan      | 1,910,025 | 126.3%       | 74.0%   | 83.3%   | -9.3 pp   | Overcapacity       |

*Table 1. Corridor recovery summary, 2025 vs. 2019 baseline. LF = load factor. Source: BTS T-100 International Segment Database.*

## **The Structural Collapse of China**

China is the defining story of the post-COVID transpacific corridor, with a collapse that has no historical counterpart in U.S. aviation history. Between 2019-2024, the U.S.-China corridor lost 3.96 million annual seats, a reduction in capacity of 75.9%. By 2025, the corridor grew slightly to 1.24 million passengers, representing a mere 29.1% of its 2019 volume. No other major corridor in the transpacific reached the end of 2025 with less than half of its 2019 passengers, placing the corridor in uncharted territory.

Bilateral frequency caps act as the largest constraint for the stagnating market. Following the pandemic, the bilateral aviation agreement between the two countries was in limbo, with both governments placing tight limits on the number of weekly flights between the countries. These constraints were only marginally loosened by 2024, with 50 weekly roundtrips per side becoming the standard. Even as U.S.-China load factors recovered above 80% by 2023, carriers could not add seats freely.

Another constraint is the closure of the Russian airspace for U.S. carriers. Beginning in 2022, after Russia's invasion of Ukraine, Russian government officials closed off the airspace to several Western countries in response to the issuance of economic sanctions. Chinese carriers, operating under no such restriction, continue to use Russian airspace and its shorter routing to fly to America with a sizable advantage. U.S. carriers were forced onto longer, more southern routes, adding 2-3 hours of block time, between \$15,000-\$20,000 in additional fuel costs per flight, and significant crew time implications. This asymmetry is apparent in T-100 load factor data: U.S. carriers operated at 72-78% load factors from 2024-2025, while Chinese carriers achieved 89-90% on the same corridor from the opposite direction.

The China capacity vacuum is the starting point for understanding where U.S. carriers went as alternatives. The redistribution can be seen in seat-level data comparing 2019 values to 2024:

| Carrier  | China Seats Lost | China Change | Japan Change | Korea Change |
|----------|------------------|--------------|--------------|--------------|
| United   | -610,134         | -68.5%       | +10.8%       | +31.1%       |
| Delta    | -374,235         | -66.5%       | -37.2%       | +33.4%       |
| American | -291,152         | -77.6%       | -9.0%        | -1.2%        |

*Table 2. U.S. legacy carrier capacity redistribution, 2019 to 2024. Source: BTS T-100 International Segment Database.*

United is the only carrier that grew Japan meaningfully, redirecting some of its lost China traffic into a corridor with joint-venture support provided by ANA in the Star Alliance. Delta redirected toward Korea via their partnership with Korean Air. American could not expand capacity in any market, despite a joint-venture agreement with JAL, choosing to reduce its Pacific footprint across the three markets.

### **Japan's Supply-Constrained Rebound**

Japan's corridor recovery: 92.5% of 2019 passengers by 2025, understates the strength of the market. The load factor operated at 83.2% in the past year, surpassing 2019 levels despite fewer passengers going around. Seat capacity in 2025 was 90.5% of 2019 levels, and the passengers who returned filled those seats more quickly than they had been doing before the pandemic, perhaps in part due to the rapid explosion of Japan travel on social media.

This signals a supply-constrained rebound, that had its supply reorganized to a high degree with Haneda's aggressive slot expansion in 2020. U.S. carriers unlocked the capability to route their aircraft through both of Tokyo's major airports, rather than being constricted purely to Narita. While this did not necessarily cause a massive increase in supply, it allowed for far-greater flexibility on the U.S.-Japan corridor.

United is the only U.S. legacy carrier growing capacity across the corridor in the period, adding 157,449 seats between 2019-2024. The joint venture agreement with ANA provides United with advantages that crowds out Delta in this context, while American's transpacific corridor is reeling following the retirement of several of its aircraft during the COVID period, a decision that has largely backfired.

Delta reduced its Japan capacity by 37.2% (358,019 less seats), redirecting its focus towards Korea. American has been hanging on, although facing a 9.0% reduction (39,648 seats) without a clear sign of reversal.

United's SFO-KIX route carried 78,175 passengers on 352 departures in 2025 at a load factor rate of 80.5%; a 29% passenger increase over 2024. At the time of this analysis, it operates as the only route from the U.S. mainland to a secondary Japanese gateway by a U.S. carrier. With United recently announcing further expansions to its Japan network, including an upcoming seasonal route from SFO-CTS (Sapporo), United is establishing a highly favorable position over its competitors in the island country.

### **South Korea's Normalization**

Korea presents itself as uncomplicated: it recovered, slightly exceeded 2019 volumes, and has largely stabilized. Passenger recovery reached 111% of 2019 values by 2025, with load factor sitting at 80.1% - a slight 3.2% decrease from its 83.3% counterpart pre-COVID. Delta's partnership with Korean Air, established in 2018, is positioning the carrier in an advantageous position and a 33.4% seat increase on Korea routes (+126,663 seats). United added 31.1% to Korea, while American marginally slipped downward (-1.2%).

LAX-ICN operates as the backbone of the corridor, carrying 596,517 passengers in 2025 across 2,039 total flights. JFK-ICN pulls an impressive 88.9% load factor, the highest on the corridor. These are mature and competitive routes operating at strong and highly sustainable load factors for the foreseeable future, and the normalization of the corridor means it is rather unlikely to generate significant network planning developments in both directions soon.

### **Taiwan's Overcapacity**

Taiwan tells a differing story from previous examples: it shows what happens when capacity grows faster than demand. Passenger recovery reached 126% of its 2019 value by 2025, the strongest recovery of any region observed in the study. On the other hand, seat capacity exploded to 142% of 2019 levels. What the passenger recovery statistic alone omits is the load factor; having fallen to 74.0%, down from a healthy 83.3% in 2019 – a 9.3 percentage point drop not driven by weak demand, but rather by a capacity overshoot not yet absorbed by the wider market.

SFO-TPE carried 596,172 passengers across a staggering 2,760 flights in 2025, operated primarily by EVA Air, United, China Airlines, and StarLux. LAX-TPE carried 458,781 passengers across 2,675 flights at a concerning 68.4% load factor, below the level of which most carriers consider transpacific service to be commercially viable.

Route-level T-100 data reveals that the load factor problem is heavily concentrated in California and is carrier-specific rather than a systematic issue across the corridor. EVA Air and StarLux are the primary drivers of it; United and mid-continent EVA routes tell a different story. LAX-TPE on StarLux and EVA Air carried 144,692 and 356,339 seats respectively, with the corresponding load factors coming out to 65.0% and 69.2% respectively.



Mid-continent routes function differently. With less capacity on routes originating from IAH and ORD, EVA's load factors shoot up to healthy numbers: 83.1% and 85.9% respectively. These are load factors more associated with healthily constrained supply levels. The correction, when it arrives, will most likely come through EVA frequency reductions or StarLux rationalizations on their California routes, rather than a spike in demand.

## **Hong Kong's Bifurcated Recovery**

Hong Kong reached 76.6% passenger recovery and 73.5% seat recovery in 2025, with load factor at 83.5%, above 2019's level of 80.1%. At the surface level, these metrics resemble Japan's supply-constrained recovery, with load factors above 2019 levels, while seats remaining below. The demand profile is different in a way that simple aggregation cannot fully capture.

Pre-COVID, Hong Kong was among the most valuable corridors in the region per seat. As a dominating global business force, Hong Kong established some of the highest rates of premium and business travel that filled front cabins at yields above Japan or Korea. Many multinational firms have relocated their regional headquarters away from the state, in some part due to the 2020 National Security Law.

BOS-HKG, a once-daily route operated by Cathay Pacific at 90.7% load factor, is the highest load factor on any U.S.-Asia route in the 2025 dataset. A monopoly route where the absence of competition and the loyalty of a specific traveler base has allowed for near-full utilization of the aircraft. JFK-HKG operates at 90.0%, while DFW-HKG pulls a still-impressive 88.8%. All these routes are operated exclusively by Cathay.

## **Carrier Unit Economics**

### **Widebody Fleet Divergence**

The most durable finding of the Form 41 analysis comes in the non-fuel CASM gap between American Airlines and its two primary competitors. This gap is present at the aircraft-level cost structure that P-52 data exposes and is not a product of temporary cost pressures.

Delta's A350-900 fleet operates at 6.66 cents/ASM non-fuel CASM, with a maintenance CASM of 0.605 cents – presenting itself as the most efficient transpacific widebody in the U.S. legacy fleet. These figures reflect the relatively young A350 fleet, under warranty coverage for a meaningful duration, and maintained under contracts that were negotiated before MRO costs surged post-COVID.

United's 787-9 operates with a non-fuel CASM at 7.11 cents – 0.45 cents above Delta's A350, with a maintenance cost of 0.422 cents, the lowest maintenance cost amongst any widebody observed in the dataset. The 787-9 is United's primary workhorse in the Pacific, having been deployed on 30.8 billion ASMs in 2025 across the entire Pacific fleet. The 777-300ER at 6.95 cents non-fuel CASM adds capacity on longer routes.

American's numbers operate on a completely different scale. The 787-9 at 9.39 cents non-fuel CASM is 2.28 cents above the equivalent of United; a gap that roughly translates to \$29,000-\$35,000 in additional non-fuel costs per flight on a 5,000-mile segment in the Pacific. This gap is only widened further with American's 777-200 and 777-300ER, operating at 10.3 and 10.4 cents respectively. With maintenance CASMs of 1.72 cents, four times that of United's 787-9, and three times Delta's A350, American is structurally disadvantaged with their high operating and maintenance costs, hammering their profitability across the corridor.

This is not a marginal cost difference that can be simply absorbed by pricing alone. At transpacific stage lengths, American's structural non-fuel CASM disadvantage compresses margins on every route across the region before a single passenger books their ticket. The 2025 operating margin difference – United at 8.0%, Delta at 9.0%, and American at 2.8%, further exemplifies this dynamic.

| Carrier  | Aircraft    | Non-Fuel CASM | Maint. CASM |
|----------|-------------|---------------|-------------|
| Delta    | A330-900neo | 6.44¢         | 0.646¢      |
| Delta    | A350-900    | 6.66¢         | 0.605¢      |
| United   | 777-300ER   | 6.95¢         | 0.659¢      |
| United   | 787-9       | 7.11¢         | 0.422¢      |
| United   | 777-200     | 7.52¢         | 0.966¢      |
| American | 787-9       | 9.39¢         | 1.720¢      |
| American | 777-200     | 10.30¢        | 1.720¢      |
| American | 777-300ER   | 10.40¢        | 1.720¢      |

*Table 3. 2025 Widebody Fleet Non-Fuel CASM, Form 41 Schedule P-52. Sorted by non-fuel CASM ascending.*

### **Carrier P&L and System Yield**

Form 41 P-12 data produces a three-year carrier P&L running from 2023-2025 that puts the widebody findings into context.

Delta generated a total of \$63.6B in operating revenue in 2025 with an operating margin of 9.0%. With the pattern largely consistent in years prior – 11.0% in 2023 and 9.7% in 2024 – this

suggests a structural performance rather than a one-off. United produced \$59.1B at an 8.0% profit margin, while American generated \$54.6B with a 2.8% profit margin, having been compressed down from 5.8% since 2023. United presents itself as the most resilient amongst the three in their profit margins; it is the only carrier to have increased its margins since 2023.

System yields from P-12 and T-2 produce a narrow band for all carriers: American at 17.5 cents/RPM, Delta at 18.1 cents, and United at 17.9 cents in 2025. This similarity closely reflects the competitive nature of the pricing environment on contested routes and explains why the seemingly large divergence in P&L is driven by cost rather than revenue. Delta's RASM premium: 22.9 cents, reflects the premium nature of Delta and its higher ancillary revenue and SkyMiles monetization.

| Year | Carrier  | Pax Revenue | Total Revenue | Op. Profit | Op. Margin |
|------|----------|-------------|---------------|------------|------------|
| 2025 | American | \$38,874M   | \$54,626M     | \$1,512M   | 2.8%       |
| 2025 | Delta    | \$42,345M   | \$63,569M     | \$5,747M   | 9.0%       |
| 2025 | United   | \$45,577M   | \$59,070M     | \$4,716M   | 8.0%       |
| 2024 | American | \$39,321M   | \$54,204M     | \$2,654M   | 4.9%       |
| 2024 | Delta    | \$41,757M   | \$61,790M     | \$6,012M   | 9.7%       |
| 2024 | United   | \$44,202M   | \$57,063M     | \$5,099M   | 8.9%       |
| 2023 | American | \$39,371M   | \$52,784M     | \$3,068M   | 5.8%       |
| 2023 | Delta    | \$39,786M   | \$58,230M     | \$6,377M   | 11.0%      |
| 2023 | United   | \$41,954M   | \$53,717M     | \$4,214M   | 7.8%       |

*Table 4. Carrier P&L Summary, 2023–2025. Source: Form 41 Schedule P-12.*

## **Pacific-Specific Performance**

Data provided by carrier reporting of Q1 2026 earnings shows the Pacific segment scope that cannot be captured in Form 41. United's Pacific yield of 16.77 cents/RPM reflects the premium that long and ultra-long-haul services in the transpacific provides. In Q1 2026, United operated

12.608B ASMs against 10.319B RPMs, producing a Pacific-wide load factor of 81.8% and \$1.73B in Pacific passenger revenue.

American's Pacific yield of 15.28 cents/RPM is reported directly in its Q1 2026 financial statement, while Delta does not disaggregate Pacific yields in its own. A proxy is therefore constructed from Delta's reported system yield of 21.78 cents using two benchmark multipliers, each built from the only two carriers that report a Pacific figure. The first is the Pacific-to-system ratio: 0.807 for United (16.77 cents / 20.77 cents) and 0.716 for American (15.28 cents / 21.34 cents), averaging 0.762, which applied to Delta's system yield gives 16.59 cents. The second is the Pacific-to-domestic ratio: 0.740 for United (16.77 cents / 22.67 cents) and 0.666 for American (15.28 cents / 22.95 cents), averaging 0.703; applied to the same system-yield anchor, this produces a more conservative 15.31 cents. The midpoint of the two, 15.95 cents, is adopted as the proxy. This places Delta squarely above American, reflecting its stronger premium mix and the increasing commercial value provided by the Korean Air joint venture, and below United, consistent with the structural advantage United holds with its San Francisco hub and tight partnership with ANA. The 15.95 cent proxy is used throughout this analysis and is flagged as a proxy where relevant.

| Carrier  | Pacific ASMs | Pacific LF | Pacific Yield | Yield Source | System Yield |
|----------|--------------|------------|---------------|--------------|--------------|
| United   | 135.7B       | 81.8%      | 16.77¢        | Reported     | ~20.77¢      |
| Delta    | 64.5B        | 84.8%      | 15.95¢        | Proxy        | 21.78¢       |
| American | 35.0B        | 84.3%      | 15.28¢        | Reported     | ~21.34¢      |

*Table 5. Q1 2026 Pacific segment metrics by carrier. Pacific ASMs from Form 41 T-1 (full year 2025). Pacific yield and load factor from Q1 2026 earnings releases; Delta yield is an indexed estimate (see Limitations)*

## The Case for Fukuoka

### Market Context

Fukuoka is Japan's fourth largest metro by population, and the commercial hub of Kyushu, and is the closest major Japanese urban center to mainland Asia; being approximately 200 kilometers from Busan and less than 900 kilometers from Shanghai. The city's economy is supported by Kyushu's advanced manufacturing base and has become an increasingly significant startup point supported by government investment in tech industry relocation away from Tokyo.

Fukuoka Airport (FUK) processes approximately 24 million annual passengers, operating as a domestic hub for both ANA and JAL, with connections to over a dozen domestic destinations

within Japan. International service from Fukuoka is well-developed, complete with direct routes to several destinations within Asia – Seoul, Beijing, Hong Kong, Taipei, and Singapore amongst others – but direct U.S. service does not currently exist. In previous years, Delta operated direct routes to the city from Honolulu, yet this service has since been discontinued. Due to the route being primarily leisure-oriented, the weakening of the yen makes re-establishment of this route rather murky.

Fukuoka handled more than 3 million international passengers pre-COVID, and the Kyushu region’s corporate and travel bases, anchored by Toyota’s manufacturing operations in Miyawaka, semiconductor facilities in Kumamoto, and increasing installations of national defense, represent a traveler profile less in line with leisure, and less-concentrated in a single industry such as Nagoya’s intense focus in automotives.

The context of supply presents itself as a key enabler. The U.S.-Japan corridor as a whole is operating above 2019 load factors with seat capacity roughly 9.5% below 2019 levels. A new Fukuoka service would enter a market that has demonstrated the ability to absorb capacity without compromising the load factor. The alternative for a network planner is to add frequency on an existing Japan “trunk” route in a market that already operates at a healthy load factor – adding to competitive density rather than creating an entirely new catchment.

### **Route Candidates**

Four U.S. gateways are evaluated as Fukuoka destination candidates, selected on the basis of Pacific gateway status and demographic catchment compatibility with Japan leisure and business travel demand.

**Seattle (SEA):** 5,269 great circle miles, 5,585 estimated track miles at 1.06x circuitry. Seattle is a hub for Delta’s Pacific gateway and is home to a once-daily SEA-HND operated by the carrier. Seattle sits in a strong Boeing-adjacent region with a higher level of corporate travel demand for Japan. Of the four proposed routes, Seattle runs the shortest stage length.

**San Francisco (SFO):** 5,663 great circle miles, 6,003 estimated track miles. SFO is a United stronghold and is the base of its Japan operations via the ANA joint venture. The SFO-KIX route, which this analysis uses as structural context, validates SFO’s ability to support secondary Japan service at near-daily frequency. The recent addition of a winter seasonal route to Sapporo (CTS) further validates the push to expand Japan travel by United.

**Los Angeles (LAX):** 5,995 great circle miles, 6,355 estimated track miles. LAX is the leader in transpacific volume, complete with the highest frequency Japan and Korea routes in the T-100 dataset. The relatively large demographic of the Southern Californian Japanese-American community and the port of L.A.’s Japan-linked logistics industry provides a natural opportunity for Fukuoka, yet operations remain airtight all throughout LAX.

**Chicago O'Hare (ORD):** 6,692 great circle miles, 7,495 estimated track miles at 1.12x polar circuitry. ORD presents itself as the longest of the four proposed routes and the one most penalized by the closure of the Russian airspace. At nearly 7,500 miles, ORD-FUK approaches the operating limit of the 787-9 configured for transpacific service.

## **Financial Model Results**

The route simulation applies the calibrated 1.3x IOC multiplier, corrected P-52 non-fuel CASMs, Q1 2026 Pacific yields, and circuitry-adjusted track miles. Q1 2026 Pacific load factors act as the demand proxy (UA: 81.8%, DL: 84.8%). Three fuel scenarios are evaluated, and yield sensitivity is tested at three levels relative to the carrier's Pacific average: 0% (Pacific average), +8% (modest premium), and +14% (mature secondary gateway premium).

| Carrier  | Aircraft | Pacific Yield | B/E Yield | Premium Needed | Mature Gateway Target | Margin at Pac. Avg |
|----------|----------|---------------|-----------|----------------|-----------------------|--------------------|
| United   | 787-9    | 16.77¢        | 18.6¢     | +11%           | 19.1¢                 | -10.8%             |
| Delta    | A350-900 | 15.95¢        | 17.3¢     | +8%            | 18.2¢                 | -8.4%              |
| American | 787-9    | 15.28¢        | 21-23¢    | 42-50%         | --                    | -42.1%             |

*Table 6. Fukuoka carrier viability summary. Base fuel (Q1 2026 actuals). B/E yield = round-trip break-even yield at daily (365) frequency. "Mature Gateway Target" represents the upper bound of the 10–14% premium range implied by the SFO-KIX cost analysis (see IOC Validation section).*

| Carrier | Route   | Stage (mi) | Seats (LF)  | RT Fare (±0%) | RT B/E  | ±0%    | +14%  | Mod.† | Sev.†  |
|---------|---------|------------|-------------|---------------|---------|--------|-------|-------|--------|
| UA      | SFO-FUK | 5,663      | 257 (81.8%) | \$1,899       | \$2,105 | -10.8% | +2.8% | -4.1% | -19.6% |
| UA      | LAX-FUK | 5,995      | 257 (81.8%) | \$2,011       | \$2,228 | -10.8% | +2.8% | -4.1% | -19.6% |
| UA      | ORD-FUK | 6,692      | 257 (81.8%) | \$2,244       | \$2,628 | -17.1% | -2.7% | --    | --     |
| DL      | SEA-FUK | 5,269      | 306 (84.8%) | \$1,681       | \$1,822 | -8.4%  | +4.9% | -2.2% | -18.1% |

| Carrier | Route   | Stage (mi) | Seats (LF)  | RT Fare (±0%) | RT B/E  | ±0%    | +14%   | Mod.† | Sev.†  |
|---------|---------|------------|-------------|---------------|---------|--------|--------|-------|--------|
| DL      | LAX-FUK | 5,995      | 306 (84.8%) | \$1,912       | \$2,073 | -8.4%  | +4.9%  | -2.2% | -18.1% |
| AA      | LAX-FUK | 5,995      | 257 (84.3%) | \$1,832       | \$2,603 | -42.1% | -24.6% | --    | --     |

Table 7. Fukuoka route simulation, all carriers, base fuel (Q1 2026). Daily frequency (365 departures). RT Fare = round-trip implied fare at Pacific average yield. RT B/E = round-trip breakeven fare. Margin columns relative to carrier Pacific average yield. Mod. and Sev. = fuel stress at +14% yield. †Mod. = +20% fuel at the mature gateway yield (+14%); Sev. = +65% fuel at +14% yield. RT Fare shown at Pacific average yield (0%). Source: Form 41 P-52, P-12A; Q1 2026 earnings; BTS T-100.

### Yield Sensitivity Analysis

| Carrier | Route   | Yield | Adj. Yield | RT Implied | Annual Profit | Margin |
|---------|---------|-------|------------|------------|---------------|--------|
| United  | SFO-FUK | +0%   | 16.77¢     | \$1,899    | -\$7,882,403  | -10.8% |
| United  | SFO-FUK | +8%   | 18.11¢     | \$2,051    | -\$2,058,934  | -2.6%  |
| United  | SFO-FUK | +14%  | 19.12¢     | \$2,165    | +\$2,308,668  | +2.8%  |
| United  | LAX-FUK | +0%   | 16.77¢     | \$2,011    | -\$8,344,518  | -10.8% |
| United  | LAX-FUK | +8%   | 18.11¢     | \$2,172    | -\$2,179,641  | -2.6%  |
| United  | LAX-FUK | +14%  | 19.12¢     | \$2,292    | +\$2,444,017  | +2.8%  |
| Delta   | SEA-FUK | +0%   | 15.95¢     | \$1,681    | -\$6,662,871  | -8.4%  |
| Delta   | SEA-FUK | +8%   | 17.23¢     | \$1,815    | -\$307,053    | -0.4%  |
| Delta   | SEA-FUK | +14%  | 18.18¢     | \$1,916    | +\$4,459,811  | +4.9%  |
| Delta   | LAX-FUK | +0%   | 15.95¢     | \$1,912    | -\$7,580,929  | -8.4%  |
| Delta   | LAX-FUK | +8%   | 17.23¢     | \$2,065    | -\$349,360    | -0.4%  |
| Delta   | LAX-FUK | +14%  | 18.18¢     | \$2,180    | +\$5,074,316  | +4.9%  |

*Table 8. Yield sensitivity analysis, United 787-9 and Delta A350-900, key Pacific coast routes. Base fuel (Q1 2026 actuals). Costs held constant; only yield varies. +0% = Pacific average; +8% = modest premium; +14% = mature gateway premium. American excluded — structurally unviable at any yield level.*

The sensitivity table reveals a sharp inflection point. Delta approaches near-breakeven levels at just 8% of its Pacific-average yield – a threshold that may be achievable within the first year or two of a seasonal or sub-daily launch. United requires a slightly higher value of ~11% above average yields to break even, crossing into positive territory before reaching the +14% mature threshold. The fuel stress test shows that these margins evaporate at 20% elevated fuel prices, reinforcing the case for a hypothetical launch to open with sub-daily frequency that helps to preserve load factors while yields grow rather than fully committing to 365 annual departures right after introduction.

### **Route Recommendation**

SFO-FUK for United represents the most defensible launch candidate, with LAX-FUK viable by either Delta or United under tight optimization. United would hypothetically face fewer competitive hurdles with their route out of San Francisco, while simultaneously operating as a less-congested hub compared to LAX. SEA-FUK also presents itself as a possible candidate by Delta, though a lack of a joint venture agreement with a Japanese carrier would theoretically add complications to the route's addition. As such, it holds the weakest case among the three west-coast gateways. It should be mentioned that Delta's position is more nuanced, as its Pacific yield estimates are established by means of proxy.

However, LAX-FUK is still in play from a financial standpoint for United as well. Competition from both Delta and American on Japan routes will be present, though the addition of a business-oriented route connecting Los Angeles to a new Japanese gateway presents itself as a unique opportunity. While outside the scope of this analysis, LAX-KIX by United may prove to yield interesting results, as a direct competitor to JAL who is currently operating the same route.

ORD-FUK is not recommended at launch. At Pacific average yields United's margin is -17.1%; even at the mature gateway premium (+14%), the margin remains -2.7%: still negative, driven by the compounding effect of the 1.12x polar circuitry penalty and the longest stage length of the four candidates. This route is unviable at any yield level tested.

### **Limitations**

**Delta's Pacific yield is a proxy.** Delta does not disaggregate its yields by region in its Q1 2026 earnings, reporting only a system-aggregated passenger yield of 21.78 cents. The 15.95 cent figure used throughout is the midpoint of two benchmarked estimates, both anchored on that reported system yield. The upper bound applies the average Pacific-to-system ratio of United and American (0.762) to produce 16.59 cents; the lower bound applies their average Pacific-to-domestic ratio (0.703); a steeper discount, to produce 15.31 cents. The midpoint places Delta



between United and American, consistent with its market position and competitive dynamics. If Delta's true Q1 2026 Pacific yield is closer to the upper bound (16.59 cents), its Fukuoka margins rapidly approach United's; if closer to the lower bound (15.31 cents), Delta turns marginally negative at base fuel, placing it nearer American. Planners with access to Delta's revenue management actuals could resolve this uncertainty directly.

**The mature gateway premium is a reference of plausibility.** The 10-14% premium range is inferred from the cost structure: SFO-KIX must earn above the aggregated Pacific average to be profitable, but its route-level yield is not directly observable with public BTS data. Planners with access to United's revenue management data or comparable secondary Japanese route yield histories could directly resolve this uncertainty.

**Daily frequency is an aggressive launch assumption.** The simulation models 365 annual departures – true daily services – for all observed combinations. New transpacific routes may launch at lower frequencies and ramp up over a period to establish load factors and build connections. If launching at lower frequencies, annual profit figures would be lower than the simulation implies; the reported numbers are better understood under the lens of mature-operation economics rather than startup projections.

**The IOC multiplier is an industry-standard assumption.** The 1.3x value is largely consistent with published long-haul widebody cost ratios and produces total CASMs consistent with corresponding CASM-ex values reported in earnings. A broader set of validations covering multiple carriers and varying stage lengths would improve confidence in the multiplier for specific route-costing applications.

**Form 41 P-52 non-fuel CASM includes depreciation.** This creates a comparability issue between carriers who finance their fleets with differing structures. Carriers who outright own their aircraft may report higher depreciation rates than those with lease-heavy fleets with lease costs appearing elsewhere in the operating costs. The relative ranking between carriers on non-fuel CASM is directionally reliable, but absolute figures should be interpreted with the caveat that ownership structure will affect what P-52 captures.

**Chinese carrier load factors may be reflecting selection effects.** The 89-90% load factors reported for Chinese carriers on U.S.-China routes in 2024-2025 may be partially reflecting the effects of selection: where carriers under bilateral frequency restrictions are more likely operate only their highest-demand routes, causing for a mechanical elevation of the reported averages. This dynamic does not negate the Russian airspace competitive asymmetry, which is documented independently by the DOT and individual carriers, but the reported double-digit percentage point gap may be overstating the true structural disadvantage held by U.S. carriers.

## **Conclusion**

The post-COVID transpacific reorganization has produced a network that may appear familiar in aggregate, while structurally different in detail. Five corridors diverged around five different

scenarios: collapse, constraint, normalization, overcapacity, and bifurcation. The three major international U.S. carriers responded with three different approaches: United invested into Japan, Delta pivoted to Korea, while American retreated, choosing to focus more on the Atlantic and Latin American regions.

Unit economics data has shown why the divergence has proven to be durable. Delta's A350 fleet and United's 787 fleet operate transpacific service at non-fuel CASM figures of 6.66 cents and 7.11 cents respectively. As American's operations cost 9.39 cents per nonfuel ASM, it is presented with a structural disadvantage that yield management cannot offset in the long-term. This is expressed in the operating margin that American recorded in 2025: 2.8%, compared to Delta's 9.0% and United's 8.0%.

Japan's supply constraint is the network opportunity identified in this analysis. A corridor operating with above-2019 load factors and below-2019 seats signals a potentially undersupplied market. The conventional response is to add frequency on existing routes, a move that United has acted on, with added service from ORD-NRT, along with an inaugural service from SFO-CTS announced recently. Fukuoka presents itself as another alternative: another new gateway, with a new passenger base without an existing U.S. nonstop competitor.

The financial case for Fukuoka is not straightforward: no route is profitable at Pacific average yields. But models show that Delta's west-coast routes break even at approximately +8% above its Pacific average, and United's break even at +11%, both within the 10-14% premium range that the SFO-KIX cost analysis implies is achievable for a mature U.S. secondary Japan monopoly gateway. Were SFO-FUK to be added as a flight, it would be the longest flight out of Fukuoka by several hours, creating a unique dynamic. While Fukuoka holds an operational curfew from 22:00-7:00, preventing any planes from departing or arriving due to noise, the prospect of connecting the westernmost major Japanese city with the United States presents itself as an intriguing opportunity.

The broader lesson of the analysis is that the transpacific market is not a uniform entity, and network planning in the region is not primarily about which corridors are showing the best growth. Taiwan has demonstrated that growth and overcapacity are not mutually exclusive, and excessively high passenger recovery numbers can be accompanied by a deteriorating load factor with oversupply. The relevant asks follow a narrower scope: where is the supply genuinely constrained, which carriers have the financials to compete in those markets, and where does the network gap allow for an opportunity that can be supported by demand at launch yield levels? On these criteria, Fukuoka presents itself as a plausible route addition under ideal circumstances and above-average yield generation.

## **Data Appendix**

The following tables provide complete source data, fleet economics, corridor stats, and simulation output supporting the analysis in the main findings. All BTS data downloaded from the public TranStats portal. Q1 2026 carrier inputs have been sourced from publicly available investor relations documents.

### **Appendix A: Data Sources and Implementation**

Table A provides each data source, the range used, and the specific fields used, with their roles in the analysis listed. The T-2 SERVICE\_CLASS de-duplication note is a critical disclosure in methodology: T-2 records three rows per carrier-quarter-aircraft combo (service classes F, K, and Z) with identical values for ASM and RPM. Filtering to SERVICE\_CLASS = “F” before aggregation returns the corrected totals; the summing of all three rows inflates all recorded metrics by 3x and was the source of uncorrected figures in earlier drafts.

| Schedule                    | Source | Date Range | Fields Used                                                                                               | Role in Analysis                                                                                                                        |
|-----------------------------|--------|------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| T-100 International Segment | BTS    | 2019-2025  | Departures, Seats, Passengers; Carrier, Origin, Dest, Month                                               | Corridor recovery, load factor, route frequency, SFO-KIX benchmark                                                                      |
| Form 41 P-52                | BTS    | 2019-2025  | TOT_AIR_OP_EXPENSES, PILOT_FLY_OPS, TOT_DIR_MAINT, AIRFRAME_DEP, ENGINE_DEP; by carrier and aircraft type | Aircraft-level non-fuel CASM and maintenance CASM for simulation                                                                        |
| Form 41 P-12A               | BTS    | 2019-2025  | TOTAL_GALLONS, TOTAL_COST; by carrier and quarter                                                         | Fuel CASM and cost-per-gallon for system validation                                                                                     |
| Form 41 P-12                | BTS    | 2019-2025  | TRANS_REV_PAX, OP_REVENUES, OP_PROFIT_LOSS; by carrier and year                                           | System yield, RASM, operating margin for P&L summary                                                                                    |
| Form 41 T-2                 | BTS    | 2019-2025  | AVL_SEAT_MILES_320, REV_PAX_MILES_140; by carrier, quarter, aircraft type, SERVICE_CLASS                  | System ASM/RPM denominators. Note: filter SERVICE_CLASS = “F” only; F/K/Z rows are identical, summing all three inflates figures by 3×. |

| Schedule         | Source      | Date Range | Fields Used                                                                                  | Role in Analysis                                                                                |
|------------------|-------------|------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Form 41 T-1      | BTS         | 2023-2025  | AVL_SEAT_MILES_320, REV_PAX_MILES_140, REV_PAX_ENP_110; by carrier and region                | Pacific-region ASMs, RPMs, and load factor for Table 5                                          |
| Q1 2026 Earnings | UAL/DAL/AAL | Q1 2026    | Pacific ASMs, Pacific RPMs, Pacific passenger revenue, system fuel cost, system fuel gallons | Pacific yield inputs (UA: reported; AA: reported; DL: proxy), fuel CASM for simulation currency |

Table A. Data sources and pipeline. All BTS Form 41 and T-100 data are downloaded from the BTS TranStats portal. Q1 2026 earnings sourced from UAL IR, DAL IR, and AAL IR.

## **Appendix B: Widebody Fleet Economics Detail**

| Carrier | Aircraft    | Non-Fuel CASM (¢) | Maint. CASM (¢) | Fleet ASMs (2025) |
|---------|-------------|-------------------|-----------------|-------------------|
| DL      | A330-900neo | 6.44¢             | 0.646¢          | 25.1B             |
| DL      | A350-900    | 6.66¢             | 0.605¢          | 27.0B             |
| UA      | 777-300ER   | 6.95¢             | 0.659¢          | 18.1B             |
| UA      | 787-9       | 7.11¢             | 0.422¢          | 30.8B             |
| UA      | 777-200     | 7.52¢             | 0.966¢          | 45.4B             |
| DL      | 767-300ER   | 7.99¢             | 0.880¢          | 16.5B             |
| AA      | 787-8       | 9.55¢             | 1.710¢          | 17.8B             |
| AA      | 787-9       | 9.39¢             | 1.720¢          | 16.2B             |
| UA      | 787-8       | 11.90¢            | 4.310¢          | 5.7B              |
| UA      | 767-300ER   | 9.89¢             | 0.851¢          | 12.3B             |
| AA      | 777-200     | 10.30¢            | 1.720¢          | 25.7B             |
| AA      | 777-300ER   | 10.40¢            | 1.720¢          | 12.9B             |

Table B. Full 2025 widebody fleet economics, all carriers. Source: Form 41 P-52 (expenses) and T-2 (ASMs, SERVICE\_CLASS = “F” filter applied). Non-fuel CASM = total aircraft operating expenses / ASMs. Fleet ASMs rounded to nearest 0.1B. The UA 787-8’s elevated maintenance CASM reflects an aging sub-fleet with MRO costs above industry norms; the type is not used in the route simulation.

## **Appendix C: Corridor Recovery Dataset**

| Corridor  | Year | Passengers | Seats     | Flights | Load Factor | Pax Recovery | Seat Recovery |
|-----------|------|------------|-----------|---------|-------------|--------------|---------------|
| China     | 2019 | 4,246,573  | 5,214,403 | 20,984  | 81.4%       | 100.0%       | 100.0%        |
|           | 2020 | 448,142    | 1,237,661 | 8,440   | 36.2%       | 10.6%        | 23.7%         |
|           | 2021 | 101,476    | 655,755   | 8,218   | 15.5%       | 2.4%         | 12.6%         |
|           | 2022 | 54,015     | 443,854   | 6,264   | 12.2%       | 1.3%         | 8.5%          |
|           | 2023 | 368,875    | 449,211   | 6,636   | 82.1%       | 8.7%         | 8.6%          |
|           | 2024 | 1,036,003  | 1,258,441 | 11,961  | 82.3%       | 24.4%        | 24.1%         |
|           | 2025 | 1,236,200  | 1,465,697 | 12,758  | 84.3%       | 29.1%        | 28.1%         |
| Hong Kong | 2019 | 1,801,882  | 2,250,180 | 10,724  | 80.1%       | 100.0%       | 100.0%        |
|           | 2020 | 212,472    | 461,708   | 5,562   | 46.0%       | 11.8%        | 20.5%         |
|           | 2021 | 24,699     | 294,198   | 3,867   | 8.4%        | 1.4%         | 13.1%         |
|           | 2022 | 142,265    | 209,650   | 2,754   | 67.9%       | 7.9%         | 9.3%          |
|           | 2023 | 708,507    | 789,132   | 6,025   | 89.8%       | 39.3%        | 35.1%         |
|           | 2024 | 1,054,015  | 1,237,651 | 8,428   | 85.2%       | 58.5%        | 55.0%         |
|           | 2025 | 1,379,715  | 1,653,303 | 8,883   | 83.5%       | 76.6%        | 73.5%         |
| Japan     | 2019 | 5,360,676  | 6,584,900 | 31,313  | 81.4%       | 100.0%       | 100.0%        |
|           | 2020 | 1,142,518  | 2,305,050 | 17,843  | 49.6%       | 21.3%        | 35.0%         |
|           | 2021 | 330,398    | 1,642,602 | 18,366  | 20.1%       | 6.2%         | 24.9%         |

| Corridor    | Year | Passengers | Seats     | Flights | Load Factor | Pax Recovery | Seat Recovery |
|-------------|------|------------|-----------|---------|-------------|--------------|---------------|
|             | 2022 | 1,764,791  | 2,875,853 | 21,158  | 61.4%       | 32.9%        | 43.7%         |
|             | 2023 | 4,029,940  | 5,342,700 | 27,753  | 75.4%       | 75.2%        | 81.1%         |
|             | 2024 | 4,703,021  | 6,177,430 | 30,878  | 76.1%       | 87.7%        | 93.8%         |
|             | 2025 | 4,958,188  | 5,957,536 | 29,598  | 83.2%       | 92.5%        | 90.5%         |
|             | 2019 | 2,992,924  | 3,592,362 | 16,154  | 83.3%       | 100.0%       | 100.0%        |
| South Korea | 2020 | 811,847    | 1,642,142 | 15,570  | 49.4%       | 27.1%        | 45.7%         |
|             | 2021 | 515,233    | 1,641,106 | 19,219  | 31.4%       | 17.2%        | 45.7%         |
|             | 2022 | 1,714,691  | 2,392,021 | 19,071  | 71.7%       | 57.3%        | 66.6%         |
|             | 2023 | 2,917,437  | 3,501,958 | 18,958  | 83.3%       | 97.5%        | 97.5%         |
|             | 2024 | 3,177,569  | 3,797,246 | 19,426  | 83.7%       | 106.2%       | 105.7%        |
| Taiwan      | 2025 | 3,319,914  | 4,144,498 | 20,508  | 80.1%       | 110.9%       | 115.4%        |
|             | 2019 | 1,512,130  | 1,814,563 | 7,531   | 83.3%       | 100.0%       | 100.0%        |
|             | 2020 | 342,222    | 669,327   | 7,551   | 51.1%       | 22.6%        | 36.9%         |
|             | 2021 | 90,267     | 390,537   | 8,023   | 23.1%       | 6.0%         | 21.5%         |
|             | 2022 | 490,696    | 762,188   | 7,759   | 64.4%       | 32.5%        | 42.0%         |
|             | 2023 | 1,663,682  | 1,853,754 | 8,359   | 89.7%       | 110.0%       | 102.2%        |
|             | 2024 | 1,919,459  | 2,327,703 | 10,216  | 82.5%       | 126.9%       | 128.3%        |
|             | 2025 | 1,910,025  | 2,581,989 | 11,269  | 74.0%       | 126.3%       | 142.3%        |

Table C. Full U.S.–Asia corridor recovery dataset, 2019–2025. Source: BTS T-100 International Segment. All figures represent U.S.-origin to foreign-destination directional traffic only. Recovery indexed to 2019 = 100%.

## **Appendix D: Fukuoka Full Simulation Output**

Table D presents the complete route simulation output across all carrier-route combinations, including routes excluded from the main recommendation (ORD-FUK for UA and DL; all AA routes). Fuel stress scenarios are shown only at the +14% mature gateway yield; they are omitted for AA (structurally excluded) and ORD-FUK at +0% and +8% (not commercially relevant). Annual profit figures assume 365 annual departures at the carrier's Q1 2026 Pacific load factor. The Ann. Profit column reflects the operating surplus or deficit above total trip costs (direct operating \* 1.3x IOC) at the stated yield and fuel price.

Routes are included regardless of carrier-gateway fit. Model output for a given carrier-route combination does not imply that the route is operationally realistic for that carrier; some combinations would require a structural network shift to pursue. Planners should apply independent judgment on gateway compatibility before treating any row as a viable candidate.

| Carrier | Route   | Fuel | Yield | Adj. Yield | RT Implied | RT B/E  | Ann. Profit   | Margin |
|---------|---------|------|-------|------------|------------|---------|---------------|--------|
| UA      | SEA-FUK | Base | +0%   | 16.77¢     | \$1,767    | \$1,959 | -\$7,333,989  | -10.8% |
| UA      | SEA-FUK | Base | +8%   | 18.11¢     | \$1,909    | \$1,959 | -\$1,915,684  | -2.6%  |
| UA      | SEA-FUK | Base | +14%  | 19.12¢     | \$2,015    | \$1,959 | +\$2,148,044  | +2.8%  |
| UA      | SEA-FUK | Mod. | +14%  | 19.12¢     | \$2,015    | \$2,098 | -\$3,178,066  | -4.1%  |
| UA      | SEA-FUK | Sev. | +14%  | 19.12¢     | \$2,015    | \$2,410 | -\$15,161,815 | -19.6% |
| UA      | SFO-FUK | Base | +0%   | 16.77¢     | \$1,899    | \$2,105 | -\$7,882,403  | -10.8% |
| UA      | SFO-FUK | Base | +8%   | 18.11¢     | \$2,051    | \$2,105 | -\$2,058,934  | -2.6%  |
| UA      | SFO-FUK | Base | +14%  | 19.12¢     | \$2,165    | \$2,105 | +\$2,308,668  | +2.8%  |
| UA      | SFO-FUK | Mod. | +14%  | 19.12¢     | \$2,165    | \$2,254 | -\$3,415,713  | -4.1%  |

| Carrier | Route   | Fuel | Yield | Adj.<br>Yield | RT<br>Implied | RT<br>B/E | Ann. Profit   | Margin |
|---------|---------|------|-------|---------------|---------------|-----------|---------------|--------|
| UA      | SFO-FUK | Sev. | +14%  | 19.12¢        | \$2,165       | \$2,590   | -\$16,295,570 | -19.6% |
| UA      | LAX-FUK | Base | +0%   | 16.77¢        | \$2,011       | \$2,228   | -\$8,344,518  | -10.8% |
| UA      | LAX-FUK | Base | +8%   | 18.11¢        | \$2,172       | \$2,228   | -\$2,179,641  | -2.6%  |
| UA      | LAX-FUK | Base | +14%  | 19.12¢        | \$2,292       | \$2,228   | +\$2,444,017  | +2.8%  |
| UA      | LAX-FUK | Mod. | +14%  | 19.12¢        | \$2,292       | \$2,387   | -\$3,615,963  | -4.1%  |
| UA      | LAX-FUK | Sev. | +14%  | 19.12¢        | \$2,292       | \$2,742   | -\$17,250,916 | -19.6% |
| UA      | ORD-FUK | Base | +0%   | 16.77¢        | \$2,244       | \$2,628   | -\$14,711,003 | -17.1% |
| UA      | ORD-FUK | Base | +14%  | 19.12¢        | \$2,559       | \$2,628   | -\$2,668,155  | -2.7%  |
| DL      | SEA-FUK | Base | +0%   | 15.95¢        | \$1,681       | \$1,822   | -\$6,662,871  | -8.4%  |
| DL      | SEA-FUK | Base | +8%   | 17.23¢        | \$1,815       | \$1,822   | -\$307,053    | -0.4%  |
| DL      | SEA-FUK | Base | +14%  | 18.18¢        | \$1,916       | \$1,822   | +\$4,459,811  | +4.9%  |
| DL      | SEA-FUK | Mod. | +14%  | 18.18¢        | \$1,916       | \$1,958   | -\$1,962,878  | -2.2%  |
| DL      | SEA-FUK | Sev. | +14%  | 18.18¢        | \$1,916       | \$2,263   | -\$16,413,928 | -18.1% |
| DL      | SFO-FUK | Base | +0%   | 15.95¢        | \$1,806       | \$1,958   | -\$7,161,101  | -8.4%  |



| Carrier | Route   | Fuel | Yield | Adj.<br>Yield | RT<br>Implied | RT<br>B/E | Ann. Profit   | Margin |
|---------|---------|------|-------|---------------|---------------|-----------|---------------|--------|
| DL      | SFO-FUK | Base | +8%   | 17.23¢        | \$1,951       | \$1,958   | -\$330,013    | -0.4%  |
| DL      | SFO-FUK | Base | +14%  | 18.18¢        | \$2,059       | \$1,958   | +\$4,793,303  | +4.9%  |
| DL      | SFO-FUK | Mod. | +14%  | 18.18¢        | \$2,059       | \$2,104   | -\$2,109,656  | -2.2%  |
| DL      | SFO-FUK | Sev. | +14%  | 18.18¢        | \$2,059       | \$2,433   | -\$17,641,312 | -18.1% |
| DL      | LAX-FUK | Base | +0%   | 15.95¢        | \$1,912       | \$2,073   | -\$7,580,929  | -8.4%  |
| DL      | LAX-FUK | Base | +8%   | 17.23¢        | \$2,065       | \$2,073   | -\$349,360    | -0.4%  |
| DL      | LAX-FUK | Base | +14%  | 18.18¢        | \$2,180       | \$2,073   | +\$5,074,316  | +4.9%  |
| DL      | LAX-FUK | Mod. | +14%  | 18.18¢        | \$2,180       | \$2,227   | -\$2,233,337  | -2.2%  |
| DL      | LAX-FUK | Sev. | +14%  | 18.18¢        | \$2,180       | \$2,575   | -\$18,675,554 | -18.1% |
| DL      | ORD-FUK | Base | +0%   | 15.95¢        | \$2,135       | \$2,445   | -\$14,652,872 | -14.5% |
| DL      | ORD-FUK | Base | +14%  | 18.18¢        | \$2,434       | \$2,445   | -\$526,284    | -0.5%  |
| AA      | SEA-FUK | Base | +0%   | 15.28¢        | \$1,610       | \$2,287   | -\$26,815,977 | -42.1% |
| AA      | SEA-FUK | Base | +14%  | 17.42¢        | \$1,836       | \$2,287   | -\$17,896,702 | -24.6% |
| AA      | SFO-FUK | Base | +0%   | 15.28¢        | \$1,731       | \$2,458   | -\$28,821,469 | -42.1% |

| Carrier | Route   | Fuel | Yield | Adj.<br>Yield | RT<br>Implied | RT<br>B/E | Ann. Profit   | Margin |
|---------|---------|------|-------|---------------|---------------|-----------|---------------|--------|
| AA      | SFO-FUK | Base | +14%  | 17.42¢        | \$1,973       | \$2,458   | -\$19,234,496 | -24.6% |
| AA      | LAX-FUK | Base | +0%   | 15.28¢        | \$1,832       | \$2,603   | -\$30,511,197 | -42.1% |
| AA      | LAX-FUK | Base | +14%  | 17.42¢        | \$2,089       | \$2,603   | -\$20,353,948 | -24.6% |
| AA      | ORD-FUK | Base | +0%   | 15.28¢        | \$2,045       | \$3,070   | -\$40,570,889 | -50.1% |
| AA      | ORD-FUK | Base | +14%  | 17.42¢        | \$2,331       | \$3,070   | -\$29,232,694 | -31.7% |

*Table D. Full Fukuoka simulation output, all carriers and routes. Base = Q1 2026 fuel; Mod. = +20% fuel; Sev. = +65% fuel. Fuel stress shown only at +14% yield. AA fuel stress omitted (structurally excluded). Ann. Profit = annual profit at daily frequency (365 departures). Source: Form 41 P-52, P-12A; Q1 2026 earnings; BTS T-100.*